The Free Uniform Spanning Forest Is a Factor of IID

Source Audit, Ram–Volterra Applicability, and Quantitative Causal Decoder Bounds

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Abstract

This report examines all seven TeX sources of the September 25, 2026 manuscript *The free uniform spanning forest is a factor of IID*. Its graphwise Borel factor construction rests on a dimension-independent finite-tilt response, posterior innovations, and an exact causal inverse on bounded differences around an arbitrary countable path family. We identify the precise connection with the source-only Urysohn–Volterra and Bielecki arguments in the private Ram corpus, while retaining the manuscript's essential countable-coordinate and Borel extension. We prove residual and inexact-Picard error bounds, a finite external-field parameter comparison, conditional influence-tail truncation bounds, and finite-coordinate readout guarantees. A bounded Borel counterexample shows that supremum-Lipschitz causality alone does not justify spatial truncation. The forest law, joint innovation law, and universal graph equivariance remain distinct mathematical inputs, not consequences of local integral algebra. Catalogue applicability is L4; classical mechanisms and inherited source results are not counted as new contributions. Contribution assessment: C2/5, useful source-specific quantitative decoder extensions are proved, but no significant advance on the global FUSF factor theorem or certified spatial implementation is established.

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1 Problem Space and Scope

1.1 The graphwise assertion

Let G be infinite, connected, locally finite, simple, undirected, and unweighted. Its vertex and edge sets are countable: balls about any fixed vertex are finite and exhaust the graph. The free uniform spanning forest law $\nu = \mathrm{FUSF}_G$ is the weak limit of uniform spanning trees on finite connected exhaustions, with no boundary identifications. The assigned source [1] asserts a single rule

$$\Phi(G, U, e) \in \{0, 1\}$$

on graphs with uniform independent vertex marks U and a distinguished undirected edge e. The rule is Borel, uses no distinguished vertex, commutes with all labeled graph isomorphisms, and returns the entire law FUSF_G for every fixed G. This implies a conditional graph factor under every unimodular law supported on this class. Neither a common exceptional event over all graphs nor a mass-transport argument is required.

The problem is not merely to produce correct edge marginals. It requires one joint forest sample, a measurable construction across varying graphs, and equivariance. An ordinary Borel factor may inspect infinitely many labels. It does not supply a finitary stopping radius, finite-valued input alphabet, computational complexity bound, or Bernoulli isomorphism. The source assumes no amenability, transience, uniform degree bound, or degree moment.

1.2 The probability and analytic spaces

For a countable index set I, the target space is $\{0,1\}^I$ with its product sigma-field and the path space is

$$\mathcal{C}_I = \prod_{i\in I} C([0,\infty),\mathbb{R})$$

with the product compact-open topology. The source proves a sampler for any binary law whose finite posterior tilts have a uniform supremum-norm Lipschitz constant. FUSF supplies $L = 1/2$. A second application treats translation-invariant strongly Rayleigh laws on a countable group under its regular left action, including determinantal laws of Hermitian positive contractions. This is not a theorem for every action on every countable set: the equivariant exhausting sets used in that proof rely on regularity of the action. No finite generation, amenability, or trace-class assumption is imposed there.

The analytic equation is

$$Z_i(w)(t) = w_i(t) + \int_0^t b_i(s, Z(w)(s))\,ds, \qquad 0 \le b_i \le 1. \tag{1.1}$$

The drift is jointly Borel, not necessarily product-continuous or differentiable. Its quantitative assumption is

$$\sup_i |b_i(t,y) - b_i(t,y')| \le L \sup_j |y_j - y'_j| \quad \text{when the latter supremum is finite.} \tag{1.2}$$

An IID Brownian field at a positive time is almost surely unbounded over infinitely many coordinates. Indeed, for any finite M, the probability that its first N coordinates lie in $[-M,M]$ is $p_M^N \to 0$. Thus neither the input nor the solution belongs to a common bounded-state ball. Bounded *differences* are the correct domain of the estimates.

1.3 Evidence and claim categories

The source pin is `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The existing October 8 provenance audit records that none of the selected 100 manuscripts changed at `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. This report uses that audit without recrawling the repository. Identity of source bytes is not verification of a mathematical theorem.

All seven assigned TeX files and the bibliography were read, including every displayed proof. Sections 2–4 distinguish inherited arguments from the local results of Sections 5–8. External strong-Rayleigh and determinantal facts remain cited inputs. No Lean replay, independent complete formal certification, or historical-priority claim is made. No definite gap in the assigned proof chain was identified in this reading; that bounded finding is not an unconditional certification of all external dependencies.

2 Audit of the Assigned Proof Chain

2.1 Finite response and the free exhaustion limit

In `response.tex`, Lemma `lem:tree-response` gives, for a finite connected graph with weights $c_f = e^{h_f} > 0$,

$$\partial_{h_f} p_e = \mathrm{Cov}(\xi_e, \xi_f), \qquad \sum_f |\partial_{h_f} p_e| = 2p_e(1-p_e) \le \tfrac{1}{2}. \tag{2.1}$$

Its proof uses the reduced incidence columns a_f and $D = \sum_f c_f a_f a_f^{\mathsf{T}}$. Cauchy–Binet identifies $\det D$ with the tree partition function, while determinant differentiation gives $p_e = c_e a_e^{\mathsf{T}} D^{-1} a_e$. For $f \neq e$,

$$\operatorname{Cov}(\xi_e, \xi_f) = -c_e c_f (a_e^{\mathsf{T}} D^{-1} a_f)^2 \leq 0.$$

The fixed number of edges in each finite tree makes the signed row sum zero. The diagonal variance then proves (2.1). Bridges give a zero row, not a singular exception to the response estimate. These are classical matrix-tree and transfer-current mechanisms, explicitly credited by the source.

Proposition `prop:free-limit` does not apply this cardinality identity to an infinite forest. Temporarily orienting G, its finite projection is $K_H = Q_H - P_{C_H}$, where C_H is the finite circulation space, equal to the span of finite cycles, and Q_H is the projection onto the coordinates of H. Increasing finite cycle spaces have strong projection limit P_C, with C the closed span of all finite cycles. Thus $K_H \to I - P_C$ strongly. Finite inclusion determinants converge; inclusion–exclusion produces consistent cylinder probabilities and hence a countable law. Every finite cycle has zero full-inclusion probability, giving acyclicity by countability. The proof uses the closed span of finite cycles, not an unjustified equality with all square-summable divergence-free flows. Orientation reversal only conjugates finite matrices by diagonal signs. Exhaustion independence and isomorphism invariance follow at the cylinder-law level.

2.2 Finite posterior tilts and two ordered spatial limits

For finite $S \subset I$, the source defines

$$b_{i,S}(t, y) = \frac{\int x_i \exp\{\sum_{j \in S} (y_j - t/2) x_j\} \, d\nu(x)}{\int \exp\{\sum_{j \in S} (y_j - t/2) x_j\} \, d\nu(x)}. \tag{2.2}$$

The denominator is at least $\exp\{\sum_{j \in S} \min(y_j - t/2, 0)\} > 0$. Both integrals are finite sums of cylinder probabilities. Lemma `lem:forest-response` first passes the finite-tree estimate through a free-volume limit with S, t, y fixed. This proves the same $1/2$ response constant for every finite observed set, even if that set is disconnected.

For an edge e, let $H_n(e)$ be the induced ball about its unordered endpoints and $S_n(e) = E(H_n(e))$. The canonical version is

$$b_e(t, y) = \limsup_{n \to \infty} b_{e, S_n(e)}(t, y). \tag{2.3}$$

Each finite tilt is itself the limit of finite-tree ratios on $H_m(e)$ with n fixed and $m \to \infty$. Only then is the outer limsup taken. Changing the order would define a different construction without the stated justification. The limsup supplies values on *all* fields; it is not asserted that every such outer sequence converges. Comparing the two finite tilts before taking their limsups in both directions preserves (1.2), including for unbounded fields with bounded differences. Countable limits give Borel measurability, not product continuity or a rate.

2.3 Posterior means and joint innovations

In `observation.tex`, take $X \sim \nu$ independently of the Brownian family B and set $Y_i(t) = tX_i + B_i(t)$. For finitely many observed coordinates, the likelihood is the exponential in (2.2), using $X_i^2 = X_i$. The Brownian bridges are independent of the endpoints and X, so the same posterior holds given the whole finite observation history. The increasing-history martingale converges to

$$b_i(t, Y(t)) = \mathbb{E}[X_i \mid \mathcal{H}_t]$$

for each fixed i, t, where $\mathcal{H}_t$ is the full raw observation sigma-field. The manuscript supplies an L^2 maximal-inequality proof of the required bounded-martingale convergence.

This fixed-time identity must not be silently strengthened to a common exceptional set for all real times. Instead bounded Fubini yields the martingale increment identities for

$$\widehat{W}_i(t) = Y_i(t) - \int_0^t b_i(s, Y(s))\, ds. \tag{2.4}$$

Lemma `lem:innovation` identifies the *joint* Brownian law. For finitely supported θ, an increment over length h is a Gaussian increment plus a correction bounded by $h\sum_i |\theta_i|$. Its conditional mean is zero, its conditional second moment is $h\sum_i \theta_i^2 + O(h^{3/2})$, and its third absolute moment is $O(h^{3/2})$. Conditional characteristic functions therefore differ from the Gaussian value by $O(h^{3/2})$. Subdivision into N intervals makes the accumulated error $O(N^{-1/2})$. This proves finite-vector Gaussian increments independent of the entire observation past. Continuity and countable rational evaluations determine the product Wiener law. Merely checking individual coordinates are Brownian would not prove independence.

2.4 Inverse, full law transfer, and integer readout

Lemma `lem:inverse` of `decoder.tex` proves the exact inverse under (1.2). Its Picard argument is reconstructed in Section 4. By definition, the full path family Y solves (1.1) with input $\widehat{W}$. Deterministic uniqueness gives $Y = Z(\widehat{W})$ pathwise. Borel measurability of Z then transfers the *complete joint path law* to fresh independent Brownian input W. This is why the reference coupling, which initially uses a sample X, does not make the actual sampler circular.

For integer n, the summable bound $\mathbb{P}(|B_i(n)| > \delta n) \le 2e^{-\delta^2 n/2}$ and countable intersections over i and rational $\delta > 0$ give $B_i(n)/n \to 0$ simultaneously. Consequently the everywhere-defined Borel predicate

$$\Psi_i(w) = \mathbf{1}\{\limsup_{n\to\infty} Z_i(w)(n)/n > 1/2\} \tag{2.5}$$

has joint law ν. Off the probability-one convergence event, the limsup predicate still defines a value. No real-time slope limit is needed.

2.5 Vertex labels and varying-graph equivariance

Lemma `lem:edge-noise` in `implementation.tex` splits each vertex uniform into an independent key and a private sequence of uniforms. The endpoint with the smaller key owns an edge, and the other endpoint's rank in its finite neighbor list selects a private coordinate. Simplicity and local finiteness ensure distinct edges select distinct finite slots. Conditionally on all keys these slots are independent uniforms. A dyadic Gaussian midpoint construction, with a summable refinement bound, gives a Borel map from one uniform to a continuous Brownian path. Exceptional quantile inputs and nonconvergence receive fixed defaults. Global key collisions form an invariant Borel null event on each fixed graph; the final factor is set to zero there.

The source uses only finite key orderings, not a measurable global ordering of a countable graph. Re-distinguishing at an edge selected from a finite ball is Borel. Finite-tree ratios, their ordered limits, parameterized scalar integration, Picard limits, and integer readouts all preserve the joint Borel property in (G, U, e, t). Being measurable separately for each fixed graph would not suffice. Every operation respects labeled isomorphisms, and the same rule supplies the conditional law under a unimodular random graph. These implementation arguments are independent of the local Volterra error estimates below.

2.6 Strongly Rayleigh and determinantal extension

In `strongly-rayleigh.tex`, positive external fields preserve real stability. Symmetric homogenization adds dummy coordinates to obtain a fixed total count, while retaining the original marginal. The required stability theorem and Rayleigh inequality are external inputs from Borcea–Brändén–Liggett [9]. If C is the enlarged covariance, all off-diagonal terms are nonpositive and its full row sums are zero. Discarding dummy columns gives

$$\sum_{j\in F} |C_{ij}| \le 2C_{ii} \le \tfrac{1}{2}.$$

Unlike the finite-tree case this is generally an inequality. Degenerate coordinates cause no difficulty. This response estimate has prior literature [10]; its rederivation is not a new Ram result.

For a countable group, choose finite $F_n \uparrow \Gamma$ and set $S_n(h) = hF_n$. Invariance of ν gives exact finite-tilt covariance under left translations; taking limsups preserves it everywhere. The sampler and coordinatewise uniform-to-Brownian map then give an everywhere-equivariant Borel factor. These sets need not be Følner. A Hermitian positive contraction has finite determinantal marginals that are strongly Rayleigh [9]; inclusion–exclusion identifies those marginals from inclusion determinants. Invariance of the determinantal law suffices; operator commutation is a sufficient, not necessary, condition for that invariance. The report imports no claim of finitariness or of an arbitrary-action extension from this argument.

3 Actual Ram Proofs and the Transfer Boundary

3.1 Primary private proof locators

The three corpus maps are routing documents, not theorem evidence. The direct private source most closely matching M442 is the Astra-series TR6, `RamNonlin-IE-PDE-astra/TR6-lin-non-lin/TR6-lin-non-lin.tex` [2]. Its numbering must not be confused with TR4 in the separate `RamNonlinearIE-PDE` series. The following passages were read directly.

Locator in Astra TR6	Mathematical content and M442 boundary
Lines 184–327; `eq:basicbound`, `eq:localidentity`, `eq:sourceerror`	Continuous finite-dimensional states on compact finite-measure domains; exact displacement substitution and a Lipschitz source-replacement bound. Localization preserves the support and any nullspace. Brownian time derivatives and finite Taylor closure are not supplied.
Lines 328–398; theorem "Full Urysohn variation", `eq:du`, `eq:ddu`	The derivative includes both source and observation-state terms. It requires continuous state derivatives, not merely a Borel countable drift. These smooth formulas are not applied to the limsup posterior.
Lines 652–695; `eq:contraction`, `eq:contractionerr`	An invariant ball and a contraction give uniqueness and residual control. The proof is ordinary Banach contraction; an unweighted small-LT restriction is not needed for a source-only Volterra problem.

Locator in Astra TR6	Mathematical content and M442 boundary
Lines 759–798; `eq:volterra`, `eq:bielecki`	Jointly continuous kernel, finite-dimensional continuous states, global state Lipschitz constants A, B. The weighted contraction factor is $\|\lambda\|(A/\omega + BT)$. For $B = 0$, choose $\omega > \|\lambda\|A$ on every finite interval. This is the direct causal mechanism.
Lines 800–871; `eq:firstcausal`, `eq:singbound`, `eq:causaltangent`	First-kind differentiation needs regularity and an invertible diagonal branch. Weakly singular memory needs a different estimate. Observation feedback leaves a multiplier that causality alone does not control.
Lines 1230–1382; `eq:certificate`, `eq:paramlip`, `eq:implicit`	Local Newton, approximate-inverse residual, and parameter bounds with actual Banach-space hypotheses. A finite residual is not a full function-space certificate; tail, quadrature, support, and model errors must be included.

The applications route was also checked in the actual Core50 TR43, `TR-core-pat-43/TR-core-pat-43.tex`, lines 327–430 [3], including the complete proof of `thm:residual` and `eq:gronwall`. Its residual-to-state comparison is classical Gronwall on a qualified finite-dimensional differential/algebraic branch. The warning that samples do not bound a continuous residual without a derivative or enclosure estimate is directly useful here. Its event and circuit claims are not imported into the probability argument. Astra TR6's limitations, lines 2008–2087, were also read; they explicitly withhold unmeasured speedups and universal inverse claims.

3.2 What transfers, and what must be proved again

With $\lambda = 1$, $g = w$, and a source-only kernel $h'(t, s, z) = b(s, z)$, the Volterra form matches (1.1). The exact displacement identity is simply

$$\int_0^t b_i(s, z(s))\, ds = \int_0^t b_i(t - a, z(t - a))\, da.$$

This is a change of variables, not a shortened history or a solver. The real gain is the causal integral estimate. There is no instantaneous observation-state dependence inside the integrand, so TR6's B term is zero. For contrast, $f(t) = 1 + \int_0^t f(t)\, ds$ has no finite solution at $t = 1$; calling an equation causal does not remove its observation multiplier.

Three extensions are essential before applying the private proof pattern: countably many coordinates, merely Borel time/state dependence, and paths unbounded in the coordinate supremum. Finite-dimensional continuous-state TR6 does not already prove these. The source's scalar integration and factorial iteration provide precisely the required extension. No Bochner-integrability assertion in a nonseparable ℓ^∞ space is needed, and no smooth-jet expansion is appropriate. Posterior innovations, law identification, and across-graph measurability remain separate even after this extension succeeds.

4 Exact Causal Inversion on Affine Path Fibers

4.1 A complete space around a given path

Fix $T < \infty$ and an arbitrary coordinatewise continuous input w. Let $\mathcal{X}_T$ consist of coordinatewise continuous families x with $\|x\|_T = \sup_{i,t \le T} |x_i(t)| < \infty$. It is complete: a Cauchy sequence converges

uniformly in all coordinates and times, preserving each coordinate's continuity and boundedness. We do not require the map $t \mapsto x(t)$ to be norm-continuous in ℓ^∞. The affine space $w + \mathcal{X}_T$ therefore makes sense even for an unbounded input with no common coordinate modulus of continuity.

Define $\mathcal{T}_w(z) = w + \int_0^{\cdot} b(s, z(s))\, ds$ coordinatewise. It maps this affine space into itself, since its correction to w has absolute value at most T. In the equivalent difference norm $\|x\|_{T,\omega} = \sup_{i,t\le T} e^{-\omega t}|x_i(t)|$,

$$\|\mathcal{T}_w(z) - \mathcal{T}_w(v)\|_{T,\omega} \le \frac{L}{\omega}(1 - e^{-\omega T})\, \|z - v\|_{T,\omega}\,. \tag{4.1}$$

The proof integrates $Le^{-\omega(t-s)}$ after applying (1.2). Bounded Borel scalar integrands give continuous corrections. Thus $\omega > L$ proves a unique fixed point on every finite interval; uniqueness makes these fixed points compatible as T increases. Every solution of (1.1) lies in this affine space automatically. This is a countable/Borel realization of the TR6 contraction mechanism, not a claim that Brownian paths live in its original finite-dimensional space.

4.2 The source factorial estimate and measurable inverse

The source's more informative construction begins with $Z^0 = w$ and $Z^{k+1} = \mathcal{T}_w(Z^k)$. Bounded drift gives the first increment at most t. Repeated integration yields, for every $k \ge 0$,

$$\sup_i |Z_i^{k+1}(t) - Z_i^k(t)| \le \frac{L^k t^{k+1}}{(k+1)!}, \qquad \sup_i |Z_i(t) - Z_i^k(t)| \le \frac{L^k t^{k+1}}{(k+1)!}. \tag{4.2}$$

For the second estimate start from $|Z - w| \le t$ and compare the exact solution to each iterate. It is *already in the source*; replacing it by the larger sum of all increment bounds is not an improvement. Both bounds hold uniformly on $[0, T]$ with t replaced by T. When $L = 0$, use $L^0 = 1$ and the exactness of the first update.

For completeness, the first inequality follows inductively from $d_{k+1}(t) \le L\int_0^t d_k(s)\, ds$, $d_0(t) \le t$. Summability gives uniform convergence of differences. The response bound then allows passage through each scalar integral. The same induction for any other solution proves uniqueness. The evaluation map $(t, z) \mapsto z(t)$ is continuous into the product field space. Parameterized integration of bounded Borel functions makes every rational-time coordinate of each Picard iterate Borel in w. Rational evaluations generate the path sigma-field, so the limit Z is Borel. Each iterate is causal and respects any coordinate relabeling intertwining the drifts; its limit does too. Bielecki contraction alone, without this parameter-measurability argument, would leave a needed part of the factor construction unaddressed.

5 Residual and Inexact-Iteration Extensions

The results in this section are local derived statements. Their mechanisms are classical Volterra iteration and Gronwall, adapted to the exact source domain and without differentiating w.

Theorem 5.1 (Integral residual certificate)**.** *Assume* (1.2) *and* $0 \le b_i \le 1$. *Fix* $w \in \mathcal{C}_I$ *and let* $v \in \mathcal{C}_I$. *Define its actual integral residual*

$$r_i(t) = v_i(t) - w_i(t) - \int_0^t b_i(s, v(s))\, ds.$$

Suppose $\sup_i |r_i(t)| \le \rho(t)$ *for a nonnegative measurable, locally bounded function* ρ. *Then*

$$\sup_i |v_i(t) - Z_i(w)(t)| \le \rho(t) + L\int_0^t e^{L(t-s)}\rho(s)\, ds. \tag{5.1}$$

In particular, $\rho(t) \le \rho_*$ on $[0,T]$ implies error at most $\rho_* e^{Lt}$ there.

Proof. The residual equation gives $\sup_i |v_i(t) - w_i(t)| \le \rho(t) + t$; hence the differences needed in (1.2) are finite. The error $E(t) = \sup_i |v_i(t) - Z_i(w)(t)|$ is measurable, being a countable supremum of continuous functions, and is locally bounded by $\rho(t) + 2t$. Subtract the integral equations to obtain $E \le \rho + VE$, where $(Vf)(t) = L\int_0^t f(s)\,ds$. Iterating m times gives $E \le \sum_{j=0}^{m-1} V^j\rho + V^m E$. On $[0,T]$ the final term is bounded by $\|E\|_T\,(LT)^m/m! \to 0$. For $j \ge 1$, Fubini gives $V^j\rho(t) = L^j\int_0^t (t-s)^{j-1}\rho(s)\,ds/(j-1)!$. Summing these nonnegative terms proves (5.1), including nonmonotone ρ. For constant ρ_* the sum is $\rho_* e^{Lt}$. □

Corollary 5.2 (Input, drift, and integration budget)**.** *Suppose a candidate satisfies*

$$v_i(t) = \widetilde{w}_i(t) + \int_0^t c_i(s, v(s))\,ds + d_i(t),$$

with defined measurable integrands. On $[0,T]$ assume

$$\sup_i |\widetilde{w}_i(t) - w_i(t)| \le a, \quad \sup_i |d_i(t)| \le d, \quad \sup_i |c_i(s, v(s)) - b_i(s, v(s))| \le \epsilon(s),$$

where $a, d \ge 0$ and $\epsilon \ge 0$ is integrable. Then

$$\sup_i |v_i(t) - Z_i(w)(t)| \le (a+d)e^{Lt} + \int_0^t e^{L(t-s)}\epsilon(s)\,ds. \tag{5.2}$$

For constant ϵ, its contribution is $\epsilon(e^{Lt}-1)/L$, interpreted as ϵt at $L = 0$.

Proof. The actual residual is bounded by $a + d + \int_0^t \epsilon(s)\,ds$. Insert this in (5.1) and exchange the nonnegative integrals. The kernel identity $1 + L\int_s^t e^{L(t-u)}\,du = e^{L(t-s)}$ gives (5.2). No Lipschitz condition on c is needed for this comparison of an already specified candidate. □

Remark 5.3 (Certificate versus observation). The bound on the residual or drift discrepancy is a hypothesis on the full time interval and all named coordinates. Finite time samples, an empirical mean-square error, or convergence of two quadrature rules do not establish it. If the final residual is evaluated in the original equation at the rounded candidate, it already includes the finite-solve errors; adding them again would double count. For $L = 1/2$, a constant drift allowance is amplified to $2\epsilon(e^{t/2} - 1)$, which is a finite-time estimate and not an automatically useful infinite-time readout guarantee.

Theorem 5.4 (Inexact Picard certificate)**.** *Let $v^0 = w$ and suppose continuous-coordinate iterates satisfy*

$$v^{k+1} = \mathcal{T}_w(v^k) + d^{k+1}, \qquad \sup_i |d_i^{k+1}(t)| \le \eta_{k+1}(t),$$

where each η_k is nonnegative, measurable, and locally bounded. With V as above, for every $k \ge 1$,

$$\sup_i |v_i^k(t) - Z_i(w)(t)| \le \frac{L^k t^{k+1}}{(k+1)!} + \sum_{j=0}^{k-1} (V^j\eta_{k-j})(t). \tag{5.3}$$

If each $\eta_k \le \eta$ on $[0,T]$, the extra error is at most $\eta\sum_{j=0}^{k-1}(Lt)^j/j! \le \eta e^{Lt}$.

Proof. Every iterate differs boundedly from w on a finite interval, by its defining equation and the locally bounded defect. Therefore the response estimate applies at every step. Put $E_k(t) = \sup_i |v_i^k(t) - Z_i(t)|$. Then $E_0(t) \le t$ and $E_{k+1} \le VE_k + \eta_{k+1}$. Unrolling this recurrence proves the first formula since $V^k(t) = L^k t^{k+1}/(k+1)!$. Integrating the constant η j times gives the second formula. No limit in k with fixed error is claimed to yield the exact solution. □

6 Finite External-Field Parameter Extension

This source-specific extension varies the target law, not just a coefficient in a formal local ODE. Let ν obey the finite-tilt sampling hypothesis with constant L. Fix a finite $\Lambda \subset I$ and a real field h supported on Λ, and define

$$d\nu^h(x) = \frac{\exp(\sum_{j\in\Lambda} h_j x_j)}{\int \exp(\sum_{j\in\Lambda} h_j z_j)\, d\nu(z)}\, d\nu(x). \tag{6.1}$$

This is a finite, strictly positive normalization, with no infinite partition function. Use the same exhausting sets $S_n(i)$ for all h.

Proposition 6.1 (Exact shift and decoder sensitivity)**.** *The canonical drift for ν^h satisfies, on every field,*

$$b_i^h(t,y) = b_i(t, y+h). \tag{6.2}$$

It has an everywhere Borel causal inverse Z^h with the same constant L, and its Brownian sampler has law ν^h. If h, h' have finite support, then for the same arbitrary input w,

$$\sup_i |Z_i^h(w)(t) - Z_i^{h'}(w)(t)| \le \|h - h'\|_\infty (e^{Lt} - 1). \tag{6.3}$$

At $L = 0$ the right side is zero. Equivariance holds when the finite field is transported as an additional mark; fixing a noninvariant field need not preserve the original unmarked symmetries.

Proof. For each i, all sufficiently large $S_n(i)$ contain Λ. Substitution of (6.1) into numerator and denominator of (2.2) cancels the prior normalizer and adds h_j to the observed field. Thus the finite-tilt identities agree eventually, proving (6.2) after limsup. Borel measurability and (1.2) pass through the fixed shift. The posterior and innovation proof applies to the binary law ν^h, and the deterministic inverse then transfers its full law exactly as in Section 2. This argument needs only the resulting drift response; alternatively one may discard the finitely many initial exhausting sets not containing Λ for each coordinate.

On any candidate field, (1.2) bounds $\sup_i |b_i^h(t,y) - b_i^{h'}(t,y)|$ by $L\,\|h - h'\|_\infty$. Apply Corollary 5.2 with identical inputs and no residual. Relabeling both y and h commutes with the shift and the given exhausting rule, which proves the stated equivariance. □

For FUSF this gives the deterministic bound $\|h - h'\|_\infty (e^{t/2} - 1)$ for any finite reweighting of edge indicators. It neither asserts an arbitrary infinite conductance-field extension nor converts finite-time closeness into exact agreement of infinite-time bits. In particular, the exponentially growing upper bound does not prove a small total-variation distance of the complete infinite output laws. A parameter derivative of the limsup drift has not been assumed or derived.

7 Spatial Truncation Requires Influence Control

7.1 A counterexample to a tempting inference

Proposition 7.1 (Supremum response is not product continuity)**.** *There is a Borel drift in $[0,1]^{\mathbb{N}}$ satisfying (1.2) with $L = 1/2$ whose exact decoder is not continuous for coordinatewise compact-uniform convergence of zero-starting input paths.*

Proof. Let $q(x) = \max(0, \min(1, x))$ and set $b_i(t, y) = \frac{1}{2} \limsup_{j\to\infty} q(y_j)$ for every i. It is Borel, bounded by $1/2$, and satisfies (1.2) because q is 1-Lipschitz and limsup preserves a uniform difference bound. For $w_j^N(t) = t\mathbf{1}_{j\le N}$, the candidate $Z(w^N) = w^N$ solves the equation: infinitely many tail coordinates remain zero and the drift vanishes. Uniqueness proves this identification. The inputs converge coordinatewise compact-uniformly to $w_j(t) = t$ for all j. For this limiting input the unique solution has all coordinates equal; for $0 \le t \le 2\log(3/2)$ they equal $z(t) = 2(e^{t/2} - 1)$, solving $z' = 1 + z/2$, $z(0) = 0$ and $z \le 1$. At every positive time in this interval, $z(t) > t$, so the corresponding outputs fail to converge even at a fixed coordinate. □

This is a counterexample for the general deterministic hypothesis, not an assertion that the FUSF posterior has this form. It explains why a row-sum response estimate alone cannot justify truncating infinitely many Brownian inputs or replacing the outer limsup by a chosen finite radius.

7.2 A conditional positive theorem with explicit tails

An adequate additional hypothesis is a *single* nonnegative matrix $A = (A_{ij})_{i,j\in I}$ with

$$\sup_i \sum_j A_{ij} \le L, \qquad |b_i(t, y) - b_i(t, y')| \le \sum_j A_{ij}|y_j - y'_j| \tag{7.1}$$

for all bounded coordinate differences and all relevant times. Such an A acts boundedly on ℓ^∞. It is stronger than bounding every finite covariance row by L: taking a supremum separately in each column before summing need not preserve that bound. This report does not establish (7.1) for the arbitrary-graph FUSF canonical drift.

Theorem 7.2 (Finite-set comparison with all omitted coordinates charged)**.** *Assume* (7.1) *and* $0 \le b_i \le 1$. *Fix finite* $F \subset I$ *and a path family* v *with* $v_j = w_j$ *for* $j \notin F$. *For* $i \in F$ *suppose*

$$v_i(t) = w_i(t) + \int_0^t c_i^F(s, v_F(s))\, ds, \qquad 0 \le c_i^F \le 1,$$

where these integrals exist. Suppose verified measurable bounds $|c_i^F(s, v_F(s)) - b_i(s, v(s))| \le \epsilon_i^F(s)$ *are given. Set* $g_i^F = \epsilon_i^F$ *in* F *and* $g_i^F = 1$ *outside* F, *and assume* $\sup_{i,s\le T} g_i^F(s) < \infty$. *Then*

$$|v_i(t) - Z_i(w)(t)| \le \int_0^t \left[e^{(t-s)A} g^F(s)\right]_i ds, \qquad t \le T. \tag{7.2}$$

If $\epsilon_i^F(s) \le \epsilon$, *the right side is at most*

$$\epsilon \frac{e^{Lt} - 1}{L} + \sum_{k=0}^{\infty} \frac{t^{k+1}}{(k+1)!} \sum_{j\notin F} (A^k)_{ij}, \tag{7.3}$$

with the usual $L = 0$ *convention in the first term.*

Proof. Both $v - w$ and $Z - w$ are bounded in absolute value by t, so all differences are bounded. Write $E_i(t) = |v_i(t) - Z_i(t)|$. Inside F, subtracting the equations and using (7.1) gives $E_i(t) \le \int_0^t [(AE(s))_i + g_i^F(s)]\, ds$. Outside F, $E_i(t) = |\int_0^t b_i(s, Z(s))\, ds| \le t$ gives the same inequality. All sums and integrals have nonnegative majorants. Iterating this inequality m times leaves a remainder bounded by $2T(LT)^m/m!$, which tends to zero. Tonelli and the norm-convergent nonnegative matrix exponential series identify the remaining terms with (7.2). The estimate $g^F \le \epsilon\mathbf{1} + \mathbf{1}_{I\setminus F}$, together with $A\mathbf{1} \le L\mathbf{1}$, gives (7.3) by termwise integration. □

Corollary 7.3 (Finite-coordinate convergence and finite-range rates)**.** *Suppose $F_N \uparrow I$ and candidates as above have uniformly bounded g^N, with $g_i^N(s) \to 0$ for each fixed i for almost every $s \in [0,T]$. Then $v_i^N \to Z_i(w)$ uniformly on $[0,T]$ for each fixed coordinate. If, in addition, $(A^k)_{ij} = 0$ for $j \notin F$ and $k < r$, the tail in (7.3) is bounded by*

$$\sum_{k=r}^{\infty} \frac{L^k t^{k+1}}{(k+1)!}. \tag{7.4}$$

For example the vanishing condition holds when A has one-step range in a specified interaction graph and i is at least r steps from $I \setminus F$.

Proof. The entries of e^{uA} increase with $u \geq 0$ and its ith row sums to at most e^{LT} on this time interval. Thus the supremum over $t \leq T$ of (7.2) is bounded by $\int_0^T \sum_j (e^{TA})_{ij} g_j^N(s)\, ds$. The measure on $I \times [0,T]$ with these summable row weights is finite. Dominated convergence makes this bound vanish. Under the additional vanishing condition, discard the zero terms and bound each remaining row sum of A^k by L^k, giving (7.4). □

The finite equations themselves have Borel causal solutions if their c^F are jointly Borel with a finite uniform Lipschitz constant, by the same finite-coordinate Picard proof. This provides existence when needed, but does not establish the discrepancies ϵ_i^F. Those must control both omitted-field and finite-volume posterior errors in the actual construction. Even using the exact infinite drift at an extended candidate would not constitute a finite-information algorithm. If F or its completion is selected by an external enumeration, graph equivariance also requires a separate check. Source finite-ball formulas are Borel but have no quantitative convergence modulus in general.

8 Finite Readouts and Asymptotic Law Preservation

Theorem 8.1 (Finite-coordinate readout budget)**.** *Let W have product Wiener law and let $Z(W)$ be the source sampler with target binary law ν. Fix integer $n \geq 1$ and finite $J \subset I$. Suppose an approximate evaluation $v(W)$ satisfies, on an event of probability at least $1-\alpha$,*

$$|v_i(W)(n) - Z_i(W)(n)| \leq e_i < n/2, \qquad i \in J,$$

with deterministic $e_i \geq 0$. Put $\widehat{X}_i = \mathbf{1}\{v_i(W)(n)/n > 1/2\}$. Then its law on J differs in total variation from ν_J by at most

$$\min\left\{1,\ \alpha + 2\sum_{i\in J} \exp\left[-\frac{(n/2 - e_i)^2}{2n}\right]\right\}. \tag{8.1}$$

No independence among the approximation errors is required.

Proof. Couple the target with $Z(W)$ by its integer-slope readout. The joint law of this pair is the joint law of (Y, X) in the reference observation model, since the slope map is Borel and recovers X there. On the successful approximation event, a wrong bit forces $|Y_i(n) - nX_i| \geq n/2 - e_i$. Since $Y_i(n) - nX_i = B_i(n)$ has variance n, the two-sided Gaussian tail is at most the corresponding term in (8.1). A union bound plus the failure probability bounds the probability that either vector differs. The coupling inequality bounds total variation by this probability, also at most one. □

For the exact kth Picard iterate and the FUSF constant $L = 1/2$, one may insert the inherited, pathwise uniform bound

$$e_i = \frac{2^{-k} n^{k+1}}{(k+1)!}. \tag{8.2}$$

This gives a certified finite-time and finite-iteration error statement *with exact drift evaluation.* It does not make that evaluation spatially finite. Inexact iteration and genuine influence-tail estimates can be inserted only after their premises have been certified. Choosing a larger n decreases observation noise but increases the deterministic error amplification; the two budgets cannot be optimized independently.

Corollary 8.2 (Sublinear error preserves the limiting bits)**.** *If, on a probability-one event, for every i, $|v_i(W)(n) - Z_i(W)(n)|/n \to 0$ along integers, then the limsup readout of v equals the source readout simultaneously at every coordinate. If v is Borel and equivariant, this replacement preserves the factor property.*

Proof. On the countable intersection of the given events and the source slope event, both normalized sequences have the same limit $X_i \in \{0, 1\}$. Their threshold predicates therefore agree. Countable limsup preserves Borel measurability, and coordinatewise thresholding preserves the assumed equivariance. □

Fixed-horizon convergence alone does not meet this premise on a growing integer horizon. Likewise, a small residual at finitely many observed edges is not a certificate for all coordinates. These readout results quantify an approximation to an already identified law; they do not replace the posterior or joint innovation proof.

9 Public Literature Check

A bounded primary-source check was performed on October 8, 2026 using public titles and mathematical terms only. It did not search or transmit private report text. Search results are not treated as an exhaustive priority census, and the pinned-source provenance audit was not repeated.

Timár's December 16, 2025 version [5], Theorem 1 and Corollary 5, proves the recurrent and invariantly amenable unimodular cases; Theorem 2(1) gives finite-valued finitary coding in the latter class. Its introduction states the full-generality FUSF question as open at that version. The assigned manuscript claims a broader graphwise Borel theorem but not those stronger finitary properties on all graphs. Lyons–Thom [6], Corollary 7.4, gives the amenable-group Bernoulli comparison in its specified action setting; neither that theorem nor its sofic coupling results alone supplies the universal graph rule here.

Nam–Sly–Zhang [7] already construct FIID Ising samples in a specified tree parameter regime through an infinite-dimensional Brownian-driven system whose law is $t\tau + B_t$. This is a close methodological antecedent, not a forest theorem. El Alaoui–Montanari [8] discuss the Gaussian observation formulation of stochastic localization. Neither the observation mechanism nor classical causal Picard iteration is claimed as new here.

The external strongly-Rayleigh inputs were checked at their primary locators: Borcea–Brändén–Liggett [9], Proposition 3.5 and Theorem 4.2, for determinantal marginals and symmetric homogenization; Alishahi–Barzegar–Zamani [10], Lemma 2.5 and its consequence, for nonnegative covariance row sums and summable absolute covariances. The full proofs of every cited external paper were not audited. These checks confirm relevant antecedents and hypotheses, not external acceptance of the assigned 2026 manuscript. No competing full-generality theorem was established by this bounded search; absence from these results is not a proof of absence in the literature.

10 Benefits, Limitations, and Further Work

10.1 What has actually been gained

The direct Ram match is an exact source-only Volterra inverse, not a generic analogy based on the word "graph". The affine-space realization makes its countable/Borel extension explicit. The new residual certificate handles rough input paths without differentiating them; the inexact-Picard formula separates inherited factorial convergence from integration and drift error. The finite-field shift identifies a genuine parameter family of target laws with a quantitative same-noise decoder comparison. The influence theorem states exactly what extra information suffices for spatial approximation, and the readout theorem carries path error to finite-dimensional distribution error. These are useful, fully stated consequences, not empirical performance claims.

The costs are equally explicit. Supremum residual control over a countable graph can be unavailable. Exponential-in-time amplification may make long-horizon bounds expensive. The finite-tree ratios can require large determinants or enumerations, and two nested spatial limits remain. A finite Taylor model does not preserve the posterior's positivity, normalization, or law unless those properties and its error are proved. A smooth finite-dimensional derivative formula is not a derivative of the canonical countable limsup. An arbitrary input enumeration or field perturbation can break equivariance. None of the tests below measures a speedup or samples the infinite FUSF.

10.2 Contribution scale and applicability

L4 describes the relevance of the private causal-integral mechanism to an important source proof step. The independent contribution scale is C1: none demonstrated; C2: a little, not significant; C3: notable or significant; C4: substantial or high; C5: very high, substantial and significant improvement or gap closure. The realized result here is C2. The source already contains the exact countable inverse, its factorial bound, posterior innovations, and global factor construction. The added estimates are classical-method, source-specific quantitative refinements. No sharp constant beyond the inherited 1/2, new global forest theorem, proved general spatial rate, finitary upgrade, or complexity advantage has been established. Report length and the importance of the target problem do not increase this score.

10.3 Concrete future problems

The most useful next theorem would bound the effect of omitted observed fields for a specified graph class and along the actual decoding paths, with a proved spatial modulus rather than a sampled covariance matrix. An alternative is a probabilistic influence envelope with a controlled failure event, which could feed (8.1) through its α term. Either route must control the inner finite-volume limit as well as the outer observed-set limit and preserve the stated order of limits.

A second problem is a certified finite-time implementation whose integration, posterior approximation, and input-path errors satisfy (5.2) or (5.3). Merely Borel time dependence does not give a convergence rate for standard sampled quadrature. One needs an integrand-specific enclosure or another proved integration scheme. A third problem is to choose horizons, iteration counts, and spatial radii so that readout errors vanish with explicit resource bounds. A fourth is to decide whether additional structure gives a finitary coding. None follows from finite-coordinate convergence or a small finite-time total-variation bound alone. Infinite external-field families and nonregular group actions similarly require new existence, response, and equivariance arguments.

A Source and Evidence Register

The assigned build directory contains the following complete TeX read set: `paper.tex`; and `sections/introduction.tex`, `sections/response.tex`, `sections/observation.tex`, `sections/decoder.tex`, `sections/implementation.tex`, and `sections/strongly-rayleigh.tex`. The bibliography `references.bib` was also read. Source locators above are the original labels, not labels invented for this report. The cited Ising companion was recognized as background, but its full source was not assigned or audited, and no theorem from it is needed for the present deductions.

The local evidence register records SHA-256 fingerprints, byte and line counts, the exact read ranges of private sources, the three routing maps, catalogue row 442, the unchanged-source provenance finding, and primary literature URLs with their limited inspection scope. The routing maps were read in full; unrelated manuscripts they mention were not thereby read or verified. Private paths and contents remain local.

B Bounded Mathematical Checks

The accompanying `evidence/checks.py` provides exact rational diagnostics for small weighted trees, bridge degeneracy, the 1/2 response constant, factorial Picard and inexact-error coefficients, finite-field tilt identities on a small law, and finite influence-matrix propagation. These tests check finite algebra underlying the displayed formulas. They are not tests of countable measurability, an infinite-volume forest sampler, arbitrary-graph spatial convergence, or a claim of novel probability theory. The executable output records the actual count and results; no inherited suite's pass count is reused.

The tests use rational weights, so derivatives with respect to log weights are checked through their exact covariance identities. Symbolic polynomial recurrences test the indexing of the factorial and inexact Picard formulas without treating numerical agreement as an analytic proof. A rational matrix-power example checks the first possible path to an omitted coordinate, while the proof of the infinite sum remains Theorem 7.2. A finite polynomial residual vanishing at chosen nodes supplies a negative control for sample-only certification.

References

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